

MEMORANDUM OF AGREEMENT

This Memorandum of Agreement (MOA) is entered into by and between:

ILOCOS SUR POLYTECHNIC STATE COLLEGE (ISPSC), a state college duly organized and existing under Philippine laws, with Main-campus and principal address at San Nicolas, Candon City, Ilocos Sur, represented herein by its President, **Dr. MARIO P. OBRERO**, hereinafter referred to as the “**ISPSC**”;

-and-

THE CITY GOVERNMENT OF CANDON, ILOCOS SUR, a local government unit duly organized and existing under the laws of the Republic of the Philippines, with office address at the Candon City Hall, San Isidro, Candon City, Ilocos Sur, represented herein by **Hon. ERIC D. SINGSON**, City Mayor, hereinafter referred to as the “**CITY OF CANDON**.”

Collectively, ISPSC and the CITY GOVERNMENT are referred to as the “**Parties**.”

-WITNESSETH, that:

WHEREAS, ISPSC, through its Office of the Vice President for Research, Development, and Extension (OVRDE), approved and implemented a research project entitled “**ESALACan: Emergency Support & Alert Lifeline Aid of Candon City Utilizing Dijkstra’s Algorithm for Route Optimization and Data Analytics**,” hereinafter referred to as the “**ARAYAT App**,” as detailed in the Approved Research Proposal attached hereto as **Annex “A”** and made an integral part of this Agreement;

WHEREAS, the **ESALACan** System is a pioneering Smart City initiative designed to provide real-time emergency reporting, optimized routing for responders using Dijkstra’s algorithm, GIS-based mapping, and descriptive data analytics to improve disaster preparedness and emergency response in Candon City;

WHEREAS, the **ESALACan** research project was conceptualized and developed by Jonathan F. Cenezan, MIT; Ryan Jay C. Cuevas, MIT; Kim Dave B. Rafael, MIT; and Fe Airam R. Dakig, MIT, all faculty members of ISPSC, whose technical expertise and intellectual contribution are hereby formally acknowledged;

WHEREAS, the project directly supports the Philippine Disaster Risk Reduction and Management Act of 2010 (R.A. No. 10121), the Data Privacy Act of 2012 (R.A. No. 10173), and the national push for Smart City development as discussed in the Philippine Institute for Development Studies (PIDS) Smart Cities report (2021);

WHEREAS, ISPSC recognizes the CITY OF CANDON as the primary collaborating and implementing agency of the **ESALACan** System and acknowledges the importance of continuous updates and enhancements to maintain the project's relevance and alignment with the City's Smart City goals;

NOW, THEREFORE, for and in consideration of the foregoing premises, the Parties hereby agree as follows:

OBJECTIVES

1. To establish a formal partnership for the utilization, operation, and enhancement of the **ESALACan** System within Candon City;
2. To define the Intellectual Property Rights and obligations of both Parties, particularly on technology transfer, updates, and the adoption of the system by other local government units (LGUs);
3. To ensure compliance with data privacy, cybersecurity, and national disaster-management standards.

OBLIGATIONS OF THE PARTIES

A. Obligations of ISPSC

1. Implement the **ESALACan** System in the City of Candon;
2. Provide technical orientation and capacity-building sessions for City Disaster Risk Reduction and Management Office (CDRRMO) personnel and other designated users;
3. Continue to update and upgrade the **ESALACan** System, including bug fixes, software patches, and functional enhancements, for the benefit of the City's long-term Smart City program, subject to the availability of research funding or mutually agreed funding arrangements;
4. Ensure that the system complies with the Data Privacy Act of 2012 (R.A. 10173), relevant cybersecurity guidelines, and applicable standards of the National Disaster Risk Reduction and Management Council (NDRRMC).

B. Obligations of the CITY OF CANDON

1. Deploy the **ESALACan** System as an official disaster management and emergency-response platform for Candon City;
2. Designate the CDRRMO or another appropriate City Office as the lead implementing unit and provide the necessary hardware, internet connectivity, and staff for system operation and maintenance;

3. Ensure that data privacy and security measures are in place consistent with R.A. 10173 and other relevant laws;

4. Be consulted and provide prior written consent in the event that ISPSC or any other party proposes to replicate, license, or adopt the **ESALACan** System for use by other LGUs or external entities, thereby protecting the City's primary implementing rights.

INTELLECTUAL PROPERTY

1. The **ESALACan** System, including all software codes, algorithms, documentation, and related intellectual property, is owned by ISPSC under Philippine copyright law and the Intellectual Property Code of the Philippines (R.A. No. 8293).

2. ISPSC grants the CITY OF CANDON three (3) years, royalty-free, exclusive, and irrevocable license to use, and operate within Candon City for public service purposes subject to renewal or purchase by the City of Candon.

TERM AND EFFECTIVITY

This MOA shall take effect upon signing and shall remain in force for three (3) years, renewable upon mutual agreement.

FUNDING

ISPSC shall cover costs related to research, system development, and initial implementation as approved in its project budget.

The CITY OF CANDON shall shoulder expenses for system maintenance, upgrades requested beyond the research scope, and operational costs after turnover.

DATA PRIVACY AND CONFIDENTIALITY

Both Parties commit to full compliance with R.A. 10173 (Data Privacy Act of 2012) and related laws to safeguard all personal and sensitive information collected and processed by the **ESALACan** System. All data shall be used solely for emergency response and disaster risk reduction purposes.

AMENDMENTS

Any modification to this MOA shall be made in writing and signed by duly authorized representatives of both Parties.

TERMINATION

Either Party may terminate this Agreement upon thirty (30) days' written notice, subject to the settlement of obligations incurred prior to termination.

MISCELLANEOUS

Nothing in this Agreement shall be construed as creating a joint venture, partnership, or employer-employee relationship between the Parties.

IN WITNESS WHEREOF, the Parties have hereunto affixed their signatures this day of 2026, at Candon City, Ilocos Sur, Philippines.

**ILOCOS SUR POLYTECHNIC
STATE COLLEGE**

**CITY GOVERNMENT OF
CANDON, ILOCOS SUR**

By:



MARIO P. OBRERO, PhD
SUC President III



HON. ERIC D. SINGSON
City Mayor



Signed in the presence of:



ROSE MARIE C. DETRAN, MSCrim
Director for Extension



ENGR. GREGORIO C. NOVIDA, JR.
Head, CDRRMO-Candon City



JOCELYN L. ABSOLOR, EdD
Vice President for Research, Development,
and Extension



MR. MARVIN A. VALDEZ
Computer Programmer II, MIS

ACKNOWLEDGEMENT

REPUBLIC OF THE PHILIPPINES)
PROVINCE OF ILOCOS SUR) ss.
CITY OF CANDON)

BEFORE ME, this day, personally appeared **Hon. ERIC D. SINGSON**, and **Dr. MARIO P. OBRERO** who are personally known to me, and to me known to be the same persons who executed the foregoing Memorandum of Agreement and they acknowledged to me that they affixed their signatures for the purpose stated in the instrument and that they executed the same as their free and voluntary act and deed and the free act and deed of the party represented by them.

WITNESS MY AND SEAL in Candon City, Ilocos Sur, Philippines
this 16th day of February, 2026.

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BOOK NO. 72
SERIES OF 2026


ATTY. EDSSEL C. ABELLA

Notary Public

Notarial Commission No. 04-2025

PTR. No. 0751397/01-05-2026 Roll No. 54259

IBP Lifetime No. 19661

MCLE Compliance No. V111-0022826/10-29-2024

Bagani Campo, Candon City, Ilocos Sur 2710